Individual Case Safety Report (ICSR)

Task 5: Complete pharmacovigilance assessment of a case study, as a Drug Safety Associate

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Objective

To perform a complete safety evaluation of the case below and prepare an ICSR, covering the four key deliverables of a Drug Safety Associate:

01	02	03	04
ADR identification	Reporting (ICSR)	Causality assessment	Classification

Case study

A 62-year-old man with gout, hypertension and stage 3 chronic kidney disease (CKD) was started on **allopurinol 300 mg once daily** by his physician on 01 Aug 2026 to prevent gout attacks. His regular medicines were amlodipine and losartan, both taken for over 2 years.

On 04 Sep 2026 (about 5 weeks later) he developed high fever and an itchy red rash that started on the face and chest and spread over most of the body, with facial swelling and swollen lymph nodes in the neck. He visited his dentist on 05 Sep 2026 for painful mouth sores, and the dentist, noticing the rash and fever, referred him urgently to hospital, where he was admitted the same day.

Blood tests showed a high eosinophil count, raised liver enzymes and worsened kidney function. Tests for viral hepatitis, EBV and CMV were negative. A skin biopsy supported a drug reaction. He was diagnosed with **Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)**. Allopurinol was stopped on 05 Sep 2026, and he was treated with oral corticosteroids. HLA-B*58:01 testing was later reported positive. He improved steadily and was discharged on 19 Sep 2026. At follow-up on 03 Oct 2026 the rash had cleared, liver enzymes had nearly normalised and kidney function had returned to his baseline.

Deliverable 01: ADR identification

Minimum criteria for a valid ICSR

Criterion	Present?	Details
Identifiable patient	Yes	Initials S.P., male, 62 years
Identifiable reporter	Yes	Dental surgeon (referring clinician)
Suspect drug	Yes	Allopurinol
Adverse event	Yes	DRESS (rash, fever, eosinophilia, liver and kidney involvement)

All four minimum criteria are met, so this is a **valid ICSR**.

Is this an adverse drug reaction?

Question	Finding
Did the event follow the drug?	Yes. Onset about 5 weeks after starting allopurinol, the typical window for DRESS (2 to 8 weeks).
Is it a known reaction?	Yes. DRESS is a recognised, serious reaction to allopurinol.
Other causes?	Viral infections excluded. Amlodipine and losartan taken for over 2 years without problems.
Improved after stopping?	Yes. Steady improvement after withdrawal and steroid treatment.
Risk factors present?	Kidney impairment, a full 300 mg starting dose, and HLA-B*58:01 positivity.

Identification: suspected ADR. **Suspect drug:** allopurinol. **Reaction:** DRESS.

Deliverable 02: Individual Case Safety Report

Structured according to ICH E2B data elements and the PvPI reporting form

CASE IDENTIFICATION

Case reference number	ICSR-IN-2026-0001 (simulated)
Report type	Spontaneous report
Initial / follow-up	Initial report, with follow-up information to 03 Oct 2026
Date first received	06 Sep 2026
Country of occurrence	India

A. PATIENT INFORMATION

Initials	S.P.
Age / gender	62 years / Male
Weight / height	72 kg / 168 cm
Relevant medical history	Gout (5 years); hypertension (8 years); CKD stage 3, eGFR about 45 mL/min. No known drug allergies.
Other relevant history	Non-smoker; occasional alcohol. No previous skin reactions.

B. SUSPECT DRUG

Name	Allopurinol tablets
Manufacturer / batch / expiry	Not known (follow-up requested)
Dose / route / frequency	300 mg / oral / once daily
Start date / stop date	01 Aug 2026 / 05 Sep 2026 (36 days)
Indication	Prevention of gout attacks (hyperuricaemia)

Action taken with suspect drug

- ☑ **Drug withdrawn**
- ☐ Dose reduced
- ☐ Dose increased
- ☐ Dose not changed
- ☐ Not applicable
- ☐ Unknown

Dechallenge / rechallenge

- ☑ **Improved after withdrawal**
- ☐ Did not improve
- ☐ Rechallenge positive
- ☐ Not rechallenged (contraindicated)

Not rechallenged: re-exposure after DRESS is contraindicated.

CONCOMITANT MEDICINES

Name	Dose / route / frequency	Start	Status	Indication
Amlodipine	5 mg oral once daily	2023	Continued	Hypertension
Losartan	50 mg oral once daily	2024	Continued	Hypertension, kidney protection

C. ADVERSE REACTION

Reported term	Fever, widespread rash, facial swelling, swollen lymph nodes, mouth sores
Coded term (MedDRA PT)	Drug reaction with eosinophilia and systemic symptoms
Date of onset	04 Sep 2026
Time to onset	35 days after first dose
Treatment given	Allopurinol stopped; oral prednisolone with gradual taper; antihistamines; skin care and fluids

RELEVANT LABORATORY RESULTS

Test	Before (Jul 2026)	Admission (05 Sep)	Follow-up (03 Oct)
Eosinophils (/cumm)	250	2,400 (high)	350
ALT (U/L)	28	210 (high)	48
Serum creatinine (mg/dL)	1.6	2.4 (high)	1.7
Atypical lymphocytes	Absent	Present	Absent
Hepatitis B/C, EBV, CMV	-	Negative	-
Skin biopsy	-	Consistent with drug reaction	-
HLA-B*58:01	-	Positive (reported later)	-

SERIOUSNESS AND OUTCOME

Seriousness criteria

☐ Death	☐ Life threatening
☑ **Hospitalisation (initial / prolonged)**	☐ Disability
☐ Congenital anomaly	☐ Other medically important

Also considered medically important: DRESS involves internal organs (liver and kidney here).

Outcome at last follow-up (03 Oct 2026)

☐ Recovered	☑ **Recovering**	☐ Not recovered
☐ Fatal	☐ Recovered with sequelae	☐ Unknown

D. REPORTER

Reporter	Dr. Zainab Umar, Dental Surgeon (BDS), Mumbai
Reporter type	Healthcare professional
Date of report	06 Sep 2026

Case narrative

A 62-year-old male patient (S.P.) with a history of gout, hypertension and chronic kidney disease stage 3 (baseline creatinine 1.6 mg/dL) was started on oral allopurinol 300 mg once daily on 01 Aug 2026 for prevention of gout attacks. Concomitant medicines were amlodipine 5 mg and losartan 50 mg daily, both taken for more than 2 years. On 04 Sep 2026 (day 35 of therapy) the patient developed high fever and a pruritic morbilliform rash starting on the face and chest and spreading to involve most of the body, with facial oedema, cervical lymphadenopathy and oral ulceration. He was referred by his dentist and hospitalised on 05 Sep 2026. Investigations showed eosinophilia (2,400/cumm), atypical lymphocytes, raised ALT (210 U/L) and raised creatinine (2.4 mg/dL). Serology for hepatitis B and C, EBV and CMV was negative. Skin biopsy was consistent with a drug reaction. The patient was diagnosed with DRESS. Allopurinol was withdrawn on 05 Sep 2026 and oral prednisolone was started. HLA-B*58:01 was later reported positive. The patient improved and was discharged on 19 Sep 2026. At follow-up on 03 Oct 2026 the rash had resolved, ALT was 48 U/L and creatinine was 1.7 mg/dL; the outcome is recovering. Allopurinol has been permanently discontinued and recorded as an allergy.

Deliverable 03: Causality assessment

WHO-UMC system

Criterion	Finding	Met?
Time relationship	Onset 35 days after starting, within the usual window for DRESS	Yes
Other explanations	Unlikely; viral causes excluded, other drugs long tolerated	Yes
Response to withdrawal	Clinically reasonable improvement after stopping	Yes
Rechallenge	Not done (contraindicated)	No
Known reaction	DRESS is well documented with allopurinol	Yes

Result: Probable / Likely. It cannot be Certain because there was no rechallenge.

Naranjo ADR probability scale

Question	Answer	Score
1. Previous conclusive reports?	Yes	+1
2. Appeared after the drug was given?	Yes	+2
3. Improved when the drug was stopped?	Yes	+1
4. Reappeared when the drug was given again?	Not done	0
5. Alternative causes?	No (infections excluded)	+2
6. Reappeared with placebo?	Do not know	0
7. Drug in toxic blood concentration?	Do not know	0
8. Dose-related?	Do not know	0
9. Similar reaction to same drug before?	No	0
10. Confirmed by objective evidence?	Yes (eosinophilia, biopsy)	+1
Total		**7**

Result: 7 = Probable. Both methods agree.

> **Causality:** Probable (WHO-UMC) / Probable, score 7 (Naranjo)

Deliverable 04: Classification

Parameter	Classification	Reason
Seriousness	**Serious**	Required hospitalisation; medically important (organ involvement)
Severity	**Severe**	Widespread rash with liver and kidney involvement; needed inpatient steroid treatment
Expectedness	**Expected (listed)**	Serious skin and hypersensitivity reactions including DRESS are listed in the allopurinol label
Type of ADR	**Type B**	Immune-mediated, not related to the drug's main action
Outcome	**Recovering**	Improving at last follow-up

Final classification: Serious, Severe, Expected, Type B; causality Probable.
Reporting priority: High. Serious case: report promptly to PvPI through an ADR Monitoring Centre. For a marketing authorisation holder, serious cases are reported to the regulator within 15 calendar days of receipt.

Drug Safety Associate review

Follow-up queries to the reporter

- Manufacturer name, batch number and expiry date of the allopurinol tablets.
- Exact prednisolone dose and taper schedule.
- Further follow-up: final outcome, and whether liver and kidney values returned fully to baseline.
- Copy of the discharge summary and the skin biopsy report.

Quality check before submission

Check	Status
Four minimum criteria present	Done
Dates consistent (start, onset, stop, discharge)	Done
Reaction coded to MedDRA	Done
Seriousness and outcome marked	Done
Narrative matches the structured fields	Done
Patient identity protected (initials only)	Done

Recommendations

- Allopurinol permanently stopped and recorded as an allergy; future gout prevention to use a different drug class, chosen by the physician.
- Patient advised to inform every doctor and dentist of this reaction.
- Learning point: in patients with kidney impairment, allopurinol should be started at a low dose and increased slowly; HLA-B*58:01 screening can identify patients at high risk.

Summary

Deliverable	Result
01 ADR identification	Valid ICSR; suspected ADR: allopurinol-induced DRESS
02 Reporting	ICSR completed with structured fields and narrative
03 Causality	Probable (WHO-UMC); Naranjo score 7 (Probable)
04 Classification	Serious, severe, expected, Type B; high reporting priority

Note: this case is hypothetical and was written for learning purposes.